

Advanced (Habanero)

Q1. Solve the system: $x + y = 4$, $x - y = 2$. What is x ?

- (A) 1
- (B) 2
- (C) 3
- (D) 4
- (E) 5

Q2. Solve the system: $2x + 3y = 7$, $x - 2y = -3$. What is y ?

- (A) -1
- (B) 0
- (C) 1
- (D) 2
- (E) 3

Q3. Solve the system: $x + 2y = 5$, $3x - 2y = 7$. What is x ?

- (A) 1
- (B) 2
- (C) 3
- (D) 4
- (E) 5

Q4. The system $x + y = 4$, $x + y = 5$ has:

- (A) One solution
- (B) No solution
- (C) Infinitely many solutions
- (D) Two solutions
- (E) Three solutions

Q5. Solve the system: $x - y = 2$, $2x + 2y = 4$. What is y ?

- (A) -1
- (B) 0
- (C) 1
- (D) 2
- (E) 3

Q6. The system $2x + 3y = 7$, $4x + 6y = 10$ has:

- (A) One solution
- (B) No solution
- (C) Infinitely many solutions
- (D) Two solutions
- (E) Three solutions

Q7. Solve the system: $x + 2y = 5$, $x + 2y = 3$. What is x ?

- (A) 1
- (B) 2
- (C) 3
- (D) No solution
- (E) Infinitely many solutions

Q8. The system $x - y = 1$, $3x - 3y = 2$ has:

- (A) One solution
- (B) No solution
- (C) Infinitely many solutions
- (D) Two solutions
- (E) Three solutions

Open Ended Questions (FRQ)

Q9. Solve the system $x + 2y = 7$, $3x - 2y = 5$ using substitution. Show all steps.

Q10. Solve the system $2x + y = 4$, $x - 2y = -3$ using elimination. Show all steps.

Chef's Tips

- Read each question carefully.
- Use the space provided for your answer.
- Show all work and explain your reasoning.